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AI-Integrated Project-Based Learning in Primary Education: A Systematic Review of Digital Literacy and Creativity

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Abstract

Background: The integration of Artificial Intelligence (AI) in education has grown rapidly, yet its implementation within the Project-Based Learning (PjBL) model at the primary school level still faces challenges.
Aims: This Systematic Literature Review (SLR) aimed to synthesize the empirical trends, effectiveness, and pedagogical roles of AI integration within the PjBL syntax, focusing specifically on its dual impact on primary school students' digital literacy and creativity.
Methods: Guided by the PRISMA 2020 protocol, this study analyzed 16 high-quality empirical articles (2021–2025) identified through ScienceDirect and Google Scholar. The selection process utilized a stringent inclusion threshold (≥ 70%) using a Quality Assessment (QA) protocol that synthesized Joanna Briggs Institute (JBI) standards with those proposed by Pingmuang.
Results: The thematic synthesis reveals that AI restructures the project lifecycle through three functional typologies: Facilitator (process support), Co-creator (collaborative partner), and Generator (content producer). Quantitative evidence demonstrates that AI-enhanced PjBL yields a large effect size (Cohen's d = 1.30) in conceptual understanding and a strong positive correlation between digital literacy and creativity (r=0.714). Critical analysis identifies two pathways: a "Virtuous Cycle" where AI fosters critical editing and cognitive scaffolding, and a "Vicious Cycle" where a lack of pedagogical guidance triggers instant dependency.
Conclusion: Primary school students possess the adaptive capacity to utilize AI effectively when supported by ethical surveillance and human-centered scaffolding. This review contributes a novel AI-PjBL Conceptual Model, offering a precise pedagogical blueprint for responsible AI integration in primary classrooms.

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INTRODUCTION

Project-Based Learning (PjBL) is fundamentally characterized by its focus on central curriculum concepts explored through constructive investigations, student autonomy, and the creation of realistic artifacts (Thomas, 2000). In the modern digital landscape, these projects serve as a fertile ground for developing 'digital literacies', a plurality of competencies that encompass cognitive, creative, and critical dimensions rather than mere technical proficiency (Belshaw, 2012). This pedagogical approach aligns with Vygotsky's sociocultural theory, positioning technology not as a passive tool, but as a cognitive mediator that scaffolds learners within their Zone of Proximal Development (ZPD) through social interaction and dialogue (Alim & Butsiarah, 2025). Within this context, AI is increasingly recognized as a 'creative partner' capable of supporting divergent thinking and aiding in the evaluation of creative outputs, thereby transforming the traditional project-based landscape (Pingmuang et al., 2025).

The integration of Artificial Intelligence (AI) within primary and secondary (K-12) education has recently emerged as a primary catalyst for global pedagogical restructuring, fundamentally aiming to equip young minds for the evolving landscape of future labor markets and academic pathways (Chiu, 2025). Rather than being treated as a mere supplementary technological add-on, generative

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AI tools are increasingly recognized as dynamic instruments capable of fostering student creativity, driving collaborative engagement, and offering real-time feedback that allows learners to visualize their cognitive processes directly inside the classroom environment. In today's knowledge society, creativity is recognized as a fundamental competence for personal and professional development, and as noted by Urmeneta & Romero (2025), the integration of AI attracts great interest in understanding how this technology can support the creative process. Despite the immense enthusiasm, Selwyn (2024) warns that the hyperbole surrounding AI often affects our ability to properly consider the long-term educational implications of this technology. Therefore, a critical approach is required to address crucial issues related to AI in education.

In the context of primary school education, the use of AI still presents complex challenges as well as opportunities. As emphasized by Aravantinos et al. (2024), the use of AI in education raises complex issues, not only in terms of its impact on learning and teaching outcomes but also regarding ethical considerations concerning personal data and the individual needs of each student. As stated by Eliaumra et al. (2024), integrating technology and interactive media into the teaching and learning process is crucial for fostering digital literacy skills, namely the ability of students to understand, analyze, and participate in the digital ecosystem effectively and responsibly. Therefore, specific empirical research regarding the use of AI in the primary school age range (4-12 years) is urgently needed to guide targeted implementation.

Active learning models have proven to be highly effective when integrated with AI technologies. For instance, Janah & Haryono (2025) demonstrated that Problem-Based Learning integrated with Artificial Intelligence (AI) and Information Technology (IT) significantly enhances elementary students' understanding compared to conventional methods. Building on this active learning paradigm, Project-Based Learning (PjBL) emerges as a highly suitable model for meaningful AI integration. Research by Viruel et al. (2025) highlights that AI-enhanced PjBL significantly outperforms regular PjBL by providing greater autonomy, responsive feedback, and adaptive personalization. Furthermore, as stated by Sutrisno et al. (2025)**,** students' readiness to adopt AI-assisted project learning is proven to be optimized when it is aligned with environmental values and local wisdom.

Despite the rapid growth of Artificial Intelligence (AI) in the educational sector, the current body of literature is characterized by a significant 'adult-learning bias,' as identified in the landmark review by Zawacki-Richter et al. (2019), which predominantly focuses on higher education contexts. Recent studies by Lee & Kwon (2024) highlight an even deeper imbalance within the K-12 spectrum: 80% of school-level AI studies are concentrated in secondary education. This bias is further reflected in current empirical PjBL-AI literature, where discussions are heavily dominated by higher education or senior high school contexts (e.g., Alim & Butsiarah, 2025; Mahmudah & Izzah, 2025; Simangunsong et al., 2023). Consequently, the specific pedagogical needs and cognitive capacities of primary school students in AI-integrated projects remain largely under-explored.

Furthermore, the rapid integration of generative AI in educational settings necessitates the establishment of comprehensive ethical frameworks to mitigate severe risks, including algorithmic bias, data privacy violations, and obscured authorship (Abuzir, 2026). From an interdisciplinary perspective, these ethical vulnerabilities highlight the urgency of proactive regulatory policies and critical AI literacy to prevent the institutionalization of misinformation and protect academic integrity among young learners (Al-kfairy et al., 2024). Addressing these systemic challenges requires authentic, K-12-specific empirical demonstrations to guide safe pedagogical implementation, rather than relying exclusively on adaptation models drawn from higher education paradigms (Crompton & Burke, 2024).

While macro-syntheses like Garzón et al. (2025) have attempted to shift focus toward early schooling, they often remain confined to generalized AI tools or administrative benefits, rather than examining operational integration within specific active learning structures like Project-Based Learning (PjBL). There is a notable deficiency in AI curricula designed for regular classroom coursework, most current implementations remain confined to extracurricular settings. Moreover, rapid reviews on generative AI, such as the work by Lo (2023), focused mostly on immediate ethical threats like plagiarism across

broad domains, rather than evaluating specific empirical interventions, such as ChatGPT paired with design platforms (e.g., Bahy & Majid, 2025), to assess the dual, simultaneous impact of AI on primary students' creativity and digital literacy.

This study establishes a rigorous scientific position by directly addressing the identified theoretical, empirical, and methodological voids. Theoretical novelty is achieved through the conceptualization of an analytical framework that categorizes AI roles into three functional typologies, Facilitator, Co-creator, and Generator, explicitly mapped across the Project-Based Learning (PjBL) syntax. This framework advances existing AIEd paradigms (e.g., Ouyang et al., 2022) by operationalizing AI not merely as a generalized tool, but as a structured pedagogical partner. Methodological novelty is provided by isolating empirical data strictly within primary education contexts, a sector currently representing only a small fraction of K-12 AI research (Lee & Kwon, 2024). This restriction deliberately eliminates both 'adult-learning bias' and 'secondary-level imbalance,' thereby offering a precise, age-appropriate, and culturally responsive blueprint that incorporates local wisdom to simultaneously evaluate AI's impact on students' creativity and digital literacy.

Based on this background, this Systematic Literature Review is guided by four main Research Questions (RQs) to deeply examine the relationships among the research variables: RQ1: What are the trends and characteristics of Artificial Intelligence (AI) utilization within the Project-Based Learning (PjBL) model at the primary school level? RQ2: To what extent is the effectiveness of AI integration within the PjBL syntax in enhancing primary school students' understanding and digital literacy? RQ3: How does AI play a role as a pedagogical support instrument in facilitating students' creativity during the project completion process? RQ4: What are the cognitive challenges, adaptation issues, and pedagogical implications faced by educators and students in implementing AI-assisted PjBL in primary education?

METHOD

Research Design

This study employed a Systematic Literature Review (SLR) method, strictly following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021). This approach ensured a structured, transparent, and reproducible process for identifying, selecting, and critically appraising relevant studies. Furthermore, the visual representation of the study selection process was generated using the PRISMA2020 flow diagram tool (Haddaway et al., 2022).

Search Strategy and Data Sources

A systematic literature search was conducted on December 15, 2025, adhering to the PRISMA 2020 guidelines to ensure transparency and reproducibility. To maximize coverage of high-impact literature, this study utilized ScienceDirect for direct access to curated peer-reviewed journals and Google Scholar as a comprehensive cross-database aggregator. This strategy was specifically designed to capture a broader spectrum of international and high-impact regional journals indexed across Scopus, Web of Science, and ERIC, thereby minimizing the risk of missing relevant literature due to closed-index restrictions (Gough et al., 2017).

To address the reviewer's suggestion for a broader search, an expanded Boolean string was applied to the Title, Abstract, and Keywords fields:

("artificial intelligence" OR "AI" OR "machine learning" OR "generative AI") AND ("project-based learning" OR "project-based instruction" OR "PjBL") AND ("digital literacy" OR "digital competence") AND ("creativity" OR "creative thinking") AND ("primary education" OR "elementary school" OR "K-12")

The search was restricted to peer-reviewed journal articles published between 2021 and 2025 to capture the most rapid and historically relevant advancements in Generative AI technology within educational frameworks.

Eligibility Criteria

To ensure the selected literature strictly aligned with the research objectives, specific inclusion and exclusion criteria were established prior to the screening phase. The parameters focused on the integration of AI within PjBL, the target demographic (primary education/K-12), and the specific learning outcomes (digital literacy and creativity). The detailed criteria are presented in Table 1.

Table 1. Inclusion and Exclusion Criteria

Criteria	Inclusion (Accepted)	Exclusion (Rejected)
Publication Timeline	Articles published within the 2021–2025 timeframe to capture the most recent and highly relevant advancements in Generative AI and modern educational technology.	Articles published before 2021, as they predominantly focus on early or conventional computer-assisted instruction rather than modern AI capabilities.
Document Type & Source	Peer-reviewed journal articles with accessible full texts retrieved from ScienceDirect and Google Scholar databases.	Non-peer-reviewed documents, undergraduate theses, dissertations, books, conference abstracts without full papers, and opinion pieces.
Population/Context	Studies focusing on K-12 education, primary education, elementary school students, and foundational early childhood learning.	Studies focusing exclusively on higher education (university/college students), adult learning, or corporate training environments.
Topic Scope & Intervention	Studies explicitly discussing the integration of Artificial Intelligence (AI) or Generative AI within a Project-Based Learning (PjBL) framework, with outcomes related to digital literacy or creativity.	Studies discussing general AI applications without PjBL syntax, PjBL implementation without AI integration, or pure technical computer science without pedagogical analysis.
Study Design	Empirical research articles (quantitative, qualitative, mixed-methods, or rigorous classroom action research) presenting primary data.	Non-empirical reports, purely theoretical papers without field implementation, or loosely related editorial essays.
Language	Articles written in English (for global context) or Indonesian (for localized implementation context).	Articles written in languages other than English or Indonesian.

Bias Analysis and Mitigation

To ensure the methodological rigor of this systematic review, potential biases were systematically identified and mitigated across three dimensions, following the best-practice guidelines for reporting systematic reviews in behavioral and educational sciences (Siddaway et al., 2019).

First, database bias was minimized by complementing the closed index database (ScienceDirect) with a comprehensive aggregator (Google Scholar), which ensured the inclusion of diverse regional empirical data that proprietary databases often omit. Second, publication bias was mitigated by strictly applying the inclusion criteria only to peer-reviewed empirical journal articles, thereby excluding non-peer-reviewed white papers, opinion pieces, and commercial editorials. Third, language bias was minimized by allowing the inclusion of both English and Indonesian articles, which prevented the exclusion of high-impact localized empirical implementation models while ensuring accurate semantic translation and analysis during the synthesis phase.

Selection Procedures

The study selection process strictly followed four PRISMA stages: identification, screening, eligibility, and inclusion, as illustrated in Figure 1.

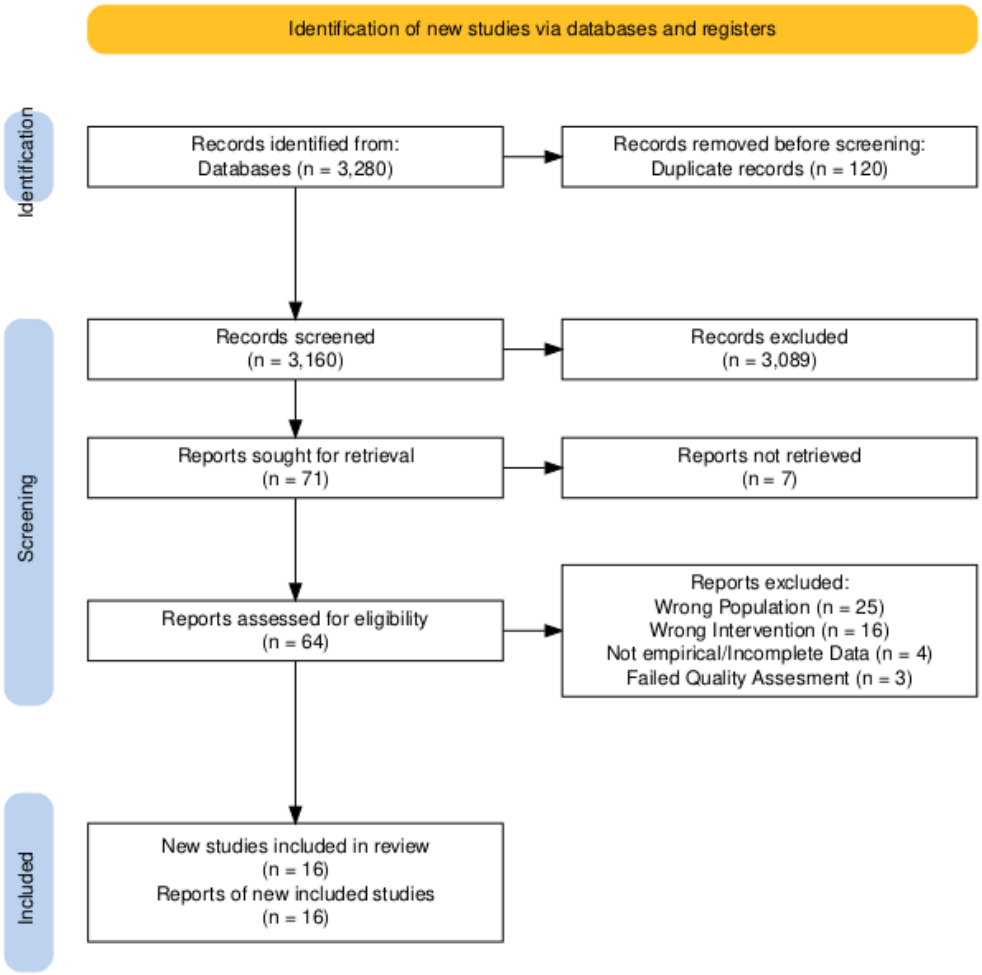


Figure 1. PRISMA 2020 Flow Diagram of the Literature Selection Process

1. Identification: A total of 3,280 records were identified across the platforms (ScienceDirect: 20, Google Scholar: 3,260). After removing 120 duplicate records, 3,160 articles remained.
2. Screening: Titles and abstracts of the 3,160 records were screened. Because Google Scholar indexes a vast amount of loosely related, non-empirical, or non-peer-reviewed documents, this phase rigorously filtered out the majority of the literature. A total of 3,089 articles were excluded for being clearly out of scope.
3. Eligibility: Of the 71 reports sought for retrieval, 7 articles could not be retrieved as full-text articles and were excluded. The remaining 64 full-text articles were assessed in-depth based on the pre-established inclusion and exclusion criteria, leading to the elimination of 45 articles. The exclusion reasons were strictly categorized: (1) Wrong population: 25 articles focused on higher education; (2) Wrong intervention: 16 articles did not integrate AI within a PjBL framework; and (3) Not empirical: 4 articles were non-empirical reports. This phase left 19 articles potentially eligible for review. Although the primary focus of this review is primary education, studies involving pre-service teachers (e.g., Simangunsong et al., 2023; Alim & Butsiarah, 2025) were intentionally retained because their structural operationalization of AI within the PjBL syntax provided foundational active-learning baselines that were directly transferable to primary education settings.
4. Inclusion: To ensure empirical validity and systematically mitigate the risk of bias, this study implemented a Quality Assessment (QA) protocol adapted from the Joanna Briggs Institute (JBI)

Critical Appraisal Checklist, adhering to the vision for critical appraisal that prioritizes internal validity and methodological conduct (Munn et al., 2022). This standardized framework, suitable for diverse educational research designs, was further integrated with the AI Ethical Use dimensions conceptualized by Pingmuang et al. (2025). While the Munn-based JBI criteria evaluated the overall rigor of the studies, the integration of Pingmuang's framework sharpened the evaluation of ethical governance, specifically requiring robust student data privacy protection, strategies to mitigate algorithmic bias, and the maintenance of explicit human oversight during AI-PjBL interventions. The ethical dimensions of the Quality Assessment (QA) protocol were further informed by the UNESCO (2023) guidance for generative AI in education, which emphasizes the protection of human agency, data privacy, and the importance of validating tools for their pedagogical appropriateness.

The scoring procedure assessed 10 critical dimensions (C1–C10), encompassing research alignment with AI-PjBL goals, clarity of PjBL syntax implementation, instrument reliability, and the inclusion of Dual Outcomes (digital literacy and creativity). Following best practices for reporting systematic reviews, each study was scored on a three-point scale: 'Yes' (1.0), 'Partial' (0.5), and 'No/Unclear'. To maintain a high standard of evidence, this review established a stringent inclusion threshold of ≥ 70% (Total Score ≥ 7.0).

This threshold was designed to filter out studies with significant methodological gaps while retaining those that provided a precise pedagogical blueprint for primary education. As detailed in Table 2, 16 articles successfully met this rigorous empirical threshold, while 3 articles were excluded due to critical methodological deficiencies or a lack of explicit pedagogical focus, thereby finalizing the sample size at 16 studies prior to data extraction.

Table 2. Quality Assessment (QA) Results

No	Author (Year)	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	Total	%	Decision
1	Anjarwati et al. (2022)	1	1	0.5	1	1	0.5	0	0.5	1	1	8.0	80%	Include
2	Yang (2022)	1	0.5	1	0.5	0.5	1	0.5	0.5	1	1	7.0	70%	Include
3	Simangunsong, et al. (2023)	1	1	1	1	1	0.5	1	0.5	1	1	9.0	90%	Include
4	Aravantinos et al. (2024)	1	0.5	0	0.5	0.5	0	0	0	0.5	0	3.0	30%	Exclude
5	Chaesar & Andayani (2024)	0.5	1	0.5	0.5	0.5	1	0	0.5	0.5	0	5.0	50%	Exclude
6	Ivcevic & Grandinetti (2024)	1	0.5	1	1	0.5	0.5	0.5	0.5	1	1	7.5	75%	Include
7	Jauhiainen & Garagorry Guerra (2024)	1	1	1	1	1	1	1	1	1	1	10.0	100%	Include
8	Kusmarni et al. (2024)	1	1	1	0.5	1	1	0.5	0.5	1	1	8.5	85%	Include
9	Viruel et al. (2025)	1	1	1	1	1	1	1	1	1	1	10.0	100%	Include
10	Samara & Kotsis (2024)	1	1	0.5	1	1	1	0.5	0.5	1	1	8.5	85%	Include
11	Zhao et al. (2024)	1	1	1	1	1	1	1	1	1	1	10.0	100%	Include
12	Alim & Butsiarah (2025)	1	1	1	1	1	1	1	0.5	1	1	9.5	95%	Include
13	Amelia & Darmawati (2025)	1	1	0.5	1	1	0.5	0.5	0.5	1	1	8.5	85%	Include

No	Author (Year)	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	Total	%	Decision
14	Bahy & Majid (2025)	1	1	1	1	0.5	1	1	0.5	1	1	9.0	90%	Include
15	Li et al. (2025)	1	0.5	1	1	1	1	1	0.5	1	1	9.0	90%	Include
16	Marcheilla et al. (2025)	1	0.5	0.5	1	0.5	0.5	0	0.5	0.5	0.5	5.5	55%	Exclude
17	Peronia & Saputro (2025)	1	1	1	1	1	1	1	1	1	1	10.0	100%	Include
18	Rustiyana (2025)	1	1	0.5	1	1	1	0.5	0.5	1	1	8.5	85%	Include
19	Sutrisno et al. (2025)	1	0.5	1	0.5	0.5	1	1	0.5	1	0.5	8.0	80%	Include

Quality Assessment Criteria (C1–C10)

1. C1: Alignment between research objectives and the AI-PjBL intervention.
2. C2: Clarity of sample description and the specific educational context.
3. C3: Validity and reliability of the data collection instruments used.
4. C4: Procedural clarity of the Project-Based Learning (PjBL) syntax implementation.
5. C5: Appropriateness of the data analysis (statistical or thematic) employed.
6. C6: Representation of voice or explicit focus on primary school learners.
7. C7: Explicit statement of ethical approval (ethics approval).
8. C8: Comprehensive discussion of research limitations or potential risk of bias.
9. C9: Conclusions that are logically supported by the presented empirical evidence.
10. C10: Explicit contribution to the development of digital literacy or creativity.

(Scoring: 1.0 = Yes; 0.5 = Partial; 0 = No/Unclear)

Data Extraction and Analysis

Following the final selection of the 16 eligible articles, data were extracted and analyzed using a thematic synthesis approach, strictly adhering to the iterative three-stage coding framework, consisting of line-by-line coding, development of descriptive themes, and generation of analytical themes proposed by Thomas & Harden (2008). This qualitative process involved coding the empirical findings to categorize the specific roles of AI within the PjBL syntax into three primary typologies: AI as a Facilitator (process support), a Co-creator (collaborative partner), and a Generator (content producer). Furthermore, the pedagogical impact of AI integration on primary students' cognitive development was systematically evaluated. Digital literacy was assessed based on students' capacity for critical evaluation and ethical use of AI outputs, while creativity was measured through the originality, flexibility, and elaboration of ideas generated during project execution.

RESULTS AND DISCUSSION

Results

Following the systematic execution of the PRISMA 2020 protocol, an initial pool of 3,280 bibliographic records was rigorously screened, culminating in the final selection of 16 empirical articles for comprehensive qualitative and thematic synthesis. To establish a transparent empirical foundation for evaluating the intersection of intelligent technologies and project-based paradigms, the structural characteristics of the included literature are systematically codified in Table 3. This literature synthesis matrix explicitly maps each study's methodological design, target sample parameters, deployed AI tools or digital platforms, identified AI functional typologies, specific nodes of PjBL syntax integration, and primary educational outcomes.

Table 3. Literature Synthesis Matrix of the Included Studies

No.	Author (Year)	Research Design	Sample Size	AI Tools/ Digital Platforms	AI Typology	PjBL Syntax Integration	Key Outcomes
1	Anjarwati et al. (2022)	Descriptive Qualitative	Elementary School	Digital Literacy Platforms	Facilitator	Contextualization & Habituation	Positive: Strengthens student character, discipline, and digital ethics.
2	Yang (2022)	Qualitative/ Framework	K-12 Education	Early AI & Machine Learning	Facilitator	Inquiry & play-based curriculum	Positive: Establishes AI literacy as an organic part of digital literacy.
3	Simangunsong et al. (2023)	Classroom Action Research	N=22, Univ. Students	AI (ChatGPT)	Co-creator	Project planning & implementation	Positive: Increases creative output significantly (up to 86.36%).
4	Ivcevic & Grandinetti (2024)	Qualitative Observational	General Education	ChatGPT & Generative AI	Co-creator	Idea generation & evaluation	Positive: AI augments human creativity across all levels (little-c to Pro-c).
5	Jauhiainen & Garagorry Guerra (2024)	Case Study	K-12 Students	Generative AI / ChatGPT	Co-creator	Idea generation & research	Positive: Acts as a brainstorming partner, enhancing creative conceptualization.
6	Kusmarni et al. (2024)	Descriptive Qualitative	Elementary School	Generative AI / ICT Platform	Generator	Content formulation & presentation	Positive: Effectively generates diverse formats for local history projects.
7	Samara & Kotsis (2024)	Experiential Intervention	Preschool	AI STEM Tools	Facilitator	Project exploration & playful learning	Positive: Significantly improves early learning of scientific concepts (magnetism).
8	Zhao et al. (2024)	Pre-Experimental	N=30, Grade 3 & 4	Generative AI	Generator	Project design & implementation	Positive: Boosts AI knowledge by 62.75% and improves technical literacy.
9	Alim & Butsiarah (2025)	Quasi-Experimental	University tudents	AI Technology (General)	Facilitator	Project designing & execution	Positive: Significantly improves complex critical thinking and analytical skills.

No.	Author (Year)	Research Design	Sample Size	AI Tools/ Digital Platforms	AI Typology	PjBL Syntax Integration	Key Outcomes
10	Amelia & Darmawati (2025)	Quasi-Experimental	Grade 5 Students	Canva AI	Generator	Project prototype & visualization	Positive: Actively generates visual outcomes, enhancing creative visualization.
11	Bahy & Majid (2025)	Quantitative Survey	N=397, K-12 Students	ChatGPT & Canva	Generator	Project creation (presentations)	Positive: Supports technical digital literacy and creative presentation skills.
12	Li et al. (2025)	Qualitative Co-design	K-12 Teachers	PBL AI Toolkit	Generator	Project implementation & creation	Neutral: Fosters creativity but highlights ethical & trust concerns.
13	Peronia & Saputro (2025)	Classroom Action Research	N=28, Grade 5	Digital Media Tools	Facilitator	Entire PjBL Syntax	Positive: 100% completeness in digital literacy improvement.
14	Viruel et al. (2025)	Quantitative Survey	N=300, Teachers	AI-driven PBL Models	Facilitator	Prototype production & evaluation	Positive: Improved student autonomy, responsive feedback, and personalization.
15	Rustiyana (2025)	Research & Development	Elementary School	Generative AI	Generator	Project prototype (Coding)	Positive: Overcomes technical barriers and boosts original thinking.
16	Sutrisno et al. (2025)	Descriptive Survey	K-12 Students	AI-based Platforms	Facilitator	Contextualization & Execution	Positive: Strengthens local wisdom and foundational competencies.

Discussion

Analysis of Research Trends and Methodological Characteristics (RQ1)

Quantitative analysis of the 16 core articles indicates a significant temporal surge in publications starting from 2023. This trend is primarily driven by the global release of generative AI tools, such as ChatGPT, which shifted the research focus from early conceptual experiments to rigorous empirical evaluations in the classroom. Geographically, contributions are predominantly clustered in Asian nations, specifically Indonesia, China, and Taiwan, reflecting a strong institutional interest in technology integration to strengthen local wisdom and 21st-century competencies.

Methodologically, the current landscape is heavily dominated by mixed-methods and quasi-experimental designs to capture the nuanced behavioral and cognitive shifts in young learners. However, a sharp data imbalance remains: 80% of school-level AI studies are concentrated in secondary education, leaving the primary sector with an urgent need for age-appropriate pedagogical blueprints. While AI integration shows potential, empirical data from the Indonesian context indicates that primary students currently possess high access to technology (92.22%) but still exhibit moderate proficiency in critical analysis and content creation (65%). This highlights an 'analytical gap' that AI-PjBL must address. Teachers and students perceive that multimodal AI-PjBL is highly effective in increasing independent learning and fostering creativity through the freedom to choose digital modes (Lee & Kwon, 2024).

Critical Synthesis: AI Typologies within PjBL Syntax (RQ2 & RQ3)

Rather than reporting studies in isolation, a consistent pattern across the literature indicates that the tactical deployment of AI systematically restructures the project lifecycle through three functional typologies. Across studies, patterns consistently show that AI does not supplant human agency but acts as a structured pedagogical partner:

1. AI as a Facilitator (Process Support): During the exploration and orientation phases, AI provides responsive feedback that guides student autonomy. Empirical evidence shows that AI-enhanced PjBL models yield a large effect size (Cohen's d = 1.30) compared to conventional PjBL.
2. AI as a Co-creator (Collaborative Partner): In the brainstorming and planning phases, AI functions as a digital 'More Knowledgeable Other' (MKO), scaffolding students within their Zone of Proximal Development (ZPD). This collaborative dynamic, rooted in Vygotsky's Social Constructivism, enables students to simplify abstract concepts and generate ideas without replacing original thought.
3. AI as a Generator (Content Producer): During the production phase, AI helps overcome technical frustrations (e.g., drawing limitations or coding barriers). This allows primary learners to focus entirely on the project's conceptual content, resulting in significant creativity gains, with one study reporting a leap from 45.45% to 86.26% proficiency.

The Dual-Edged Sword: Virtuous Scaffolding vs. Vicious Dependency (RQ4)

Critical analysis reveals that AI integration functions as a "double-edged sword" for cognitive development. The literature identifies two contradictory pathways determined by instructional design:

1. Virtuous Cycle: AI supports digital literacy when the design requires learners to evaluate, question, and revise AI-generated content. This friction between human skepticism and machine output hones analytical thinking, evidenced by a strong positive correlation between digital literacy and creativity (r = 0.714).
2. Vicious Cycle: Without precise pedagogical architecture, AI tools trigger "instant dependency". This occurs when AI delivers complete answers before students engage with the problem, leading to cognitive offloading that weakens independent problem-solving skills. This risk is exacerbated by systemic teacher unreadiness in auditing algorithmic bias. According to Celik (2023), such failures frequently stem from a deficiency in ethical-pedagogical synergy; specifically, educators possessing lower technical proficiency are more prone to over-trusting AI systems, which can compromise critical pedagogical oversight.

Narrative Integration of the Conceptual Model

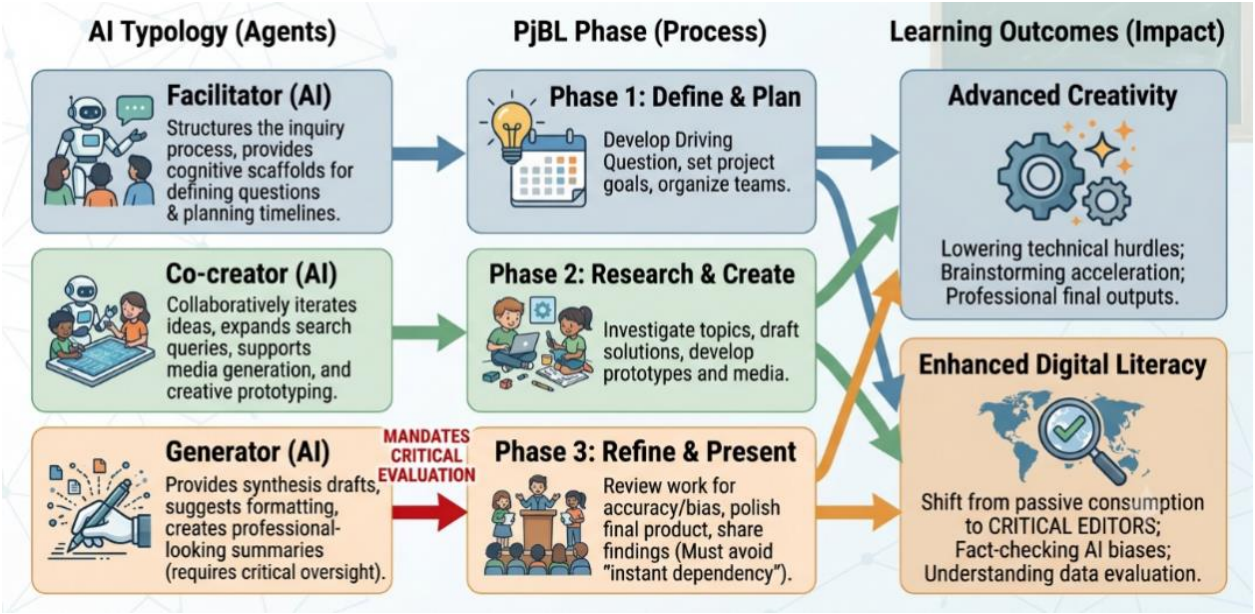


Figure 2. Conceptual Model Linking AI Typologies, PjBL Phases, and Learning Outcomes in Primary Education ((Visualized with the Assistance of Google Gemini AI based on the Author's Original Conceptual Framework)

As a final synthesis, this study proposes the AI-PjBL Conceptual Model (Figure 2), which links AI Typologies to PjBL Phases and Learning Outcomes. The model illustrates that student autonomy is optimized when AI is positioned as a Facilitator for early scaffolding, transitioning into a Co-creator for ideation, and finally a Generator for concrete visualization. This integration proves that primary school students possess the adaptive capacity to utilize generative AI effectively when supported by strict pedagogical scaffolding and ethical guidance.

Implications

The findings of this study carry significant implications for educational practitioners and curriculum designers:

1. Evolution of the Teacher's Role: Teachers must transition from being primary knowledge providers to ethical supervisors and critical facilitators. Strict scaffolding is required to prevent "instant dependency" on AI tools among young learners.
2. Early AI Literacy Frameworks: Schools need to formally integrate progressive AI literacy into the elementary curriculum. Understanding the ethics, biases, and limitations of AI must become an organic part of basic digital literacy.
3. Culturally Responsive AI Integration: To maximize engagement and readiness, educators should align AI-assisted project themes with local wisdom and environmental realities (ecoliteracy), making abstract technology more relatable for primary students.

Research Contributions

1. Theoretical Contribution: This study introduces a novel analytical framework by categorizing AI roles into three functional typologies: Facilitator, Co-creator, and Generator, specifically mapped across the PjBL syntax.
2. Practical Contribution: It provides a precise, age-appropriate pedagogical blueprint for primary educators, demonstrating that "thinking with AI" is achievable for young learners when supported by ethical surveillance and human-centered scaffolding.

Limitations

Despite the rigorous PRISMA 2020 methodology, this review acknowledges several limitations. First, although ScienceDirect and Google Scholar were utilized to capture high-impact literature across major indexes like Scopus and WoS, the exclusion of other specialized databases might have omitted niche regional studies. Second, while studies involving pre-service teachers were included as transferable pedagogical blueprints, the inherent 'adult-learning bias' in AIEd research means that findings may not fully reflect the unique socio-emotional needs of primary learners. Lastly, the synthesis highlights a systemic lack of standardized quantitative metrics for measuring AI-mediated creativity and digital literacy, forcing a reliance on diverse qualitative interpretations across the reviewed articles.

Suggestions

Based on the identified gaps, future research should prioritize:

1. Longitudinal and Neurocognitive Studies: Conducting long-term empirical work to observe the sustained impact of AI-assisted PjBL on knowledge retention, cognitive ergonomics, and reasoning-related neural functions in young children.
2. Standardized Rubrics and Child-Centric Co-design: Developing age-appropriate, quantitative rubrics for AI-mediated creativity and involving primary students directly in the co-design of AI-PjBL syntaxes to ensure the interventional tools reflect children's lived experiences.
3. Advanced Teacher Professional Development: Investigating the integration of Intelligent-TPACK and AI ethical governance within teacher training programs to equip educators with the skills to audit algorithmic bias and prevent student cognitive dependency. Future teacher training programs must move beyond tool familiarity to address instructional design specifically. As evidenced by Baskara et al. (2024), while educators successfully develop AI-integrated projects, there remains a critical need for continuous support to bridge the gap between AI's technological potential and its effective, responsible pedagogical implementation in classrooms.
4. Contextualization and Local Wisdom: Exploring how AI-PjBL models can be further tailored to local cultural heritage to enhance students' technological readiness and environmental values.

CONCLUSION

This systematic literature review concludes that the integration of Artificial Intelligence (AI) within the Project-Based Learning (PjBL) model offers a transformative pedagogical shift for primary education, provided it is guided by rigorous scaffolding. The following sections provide explicit answers to the research questions based on the synthesized empirical evidence:

1. RQ1: Research Trends and Characteristics: The analysis reveals a significant temporal surge in publications starting in 2023, directly correlating with the global release of generative AI tools like ChatGPT. Research in this field is characterized by a dominance of mixed-methods and qualitative case designs aimed at capturing nuanced behavioral shifts in young learners. Geographically, contributions are predominantly clustered in Asian nations, particularly Indonesia, China, and Taiwan, reflecting strong institutional interest in utilizing technology to strengthen local wisdom and 21st-century competencies.
2. RQ2: Effectiveness in Understanding and Digital Literacy: AI integration within the PjBL syntax is highly effective, yielding a large effect sizes (Cohen's d = 1.30) in improving conceptual understanding compared to conventional models. Regarding digital literacy, student interaction with AI triggers a paradigm shift from basic technical proficiency to rigorous critical evaluation. Students adapt by becoming "critical editors," cross-referencing AI outputs to mitigate the risks of algorithmic hallucinations and bias.
3. RQ3: AI as a Pedagogical Support for Creativity: AI functions as a robust creativity amplifier rather than a replacement for human imagination. By acting as a Co-creator during brainstorming and a Generator during production, AI lowers technical entry barriers, such as coding or complex visualization, allowing primary students to focus on conceptual depth.

Empirical data show significant gains in creative proficiency, with one study reporting a leap from 45.45% to 86.26% in project originality.

4. RQ4: Challenges and Pedagogical Implications: AI integration acts as a "double-edged sword." While it supports a "Virtuous Cycle" of cognitive scaffolding, a lack of structured guidance can trigger a "Vicious Cycle" of instant dependency and passive copying. A critical challenge is the systemic pedagogical unreadiness of teachers in auditing AI bias. The findings imply that student readiness and teacher adoption are maximized when AI projects are contextualized within local wisdom and environmental values.

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AUTHOR CONTRIBUTION STATEMENT

E.N.R.: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Writing – original draft. M.: Supervision, Validation, Writing – Review & Editing. A.S.: Supervision, Validation, Writing – review & editing. All authors have read and agreed to the published version of the manuscript.

AI DISCLOSURE STATEMENT

The authors utilized AI-assisted technologies, specifically Google Gemini, to facilitate the visual conceptualization and graphic generation of Figure 2 based on the authors' original theoretical framework. Additionally, Generative AI tools were used to improve language readability and structural formatting during the drafting process. All intellectual content, critical analysis, and the core logic of the conceptual model remain original and are the sole responsibility of the authors.

CONFLICTS OF INTEREST

The authors confirm the presence or absence of any potential conflicts of interest—financial, institutional, or personal, that could influence the conduct of this study, the analysis of data, the preparation of the manuscript, or its publication.

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